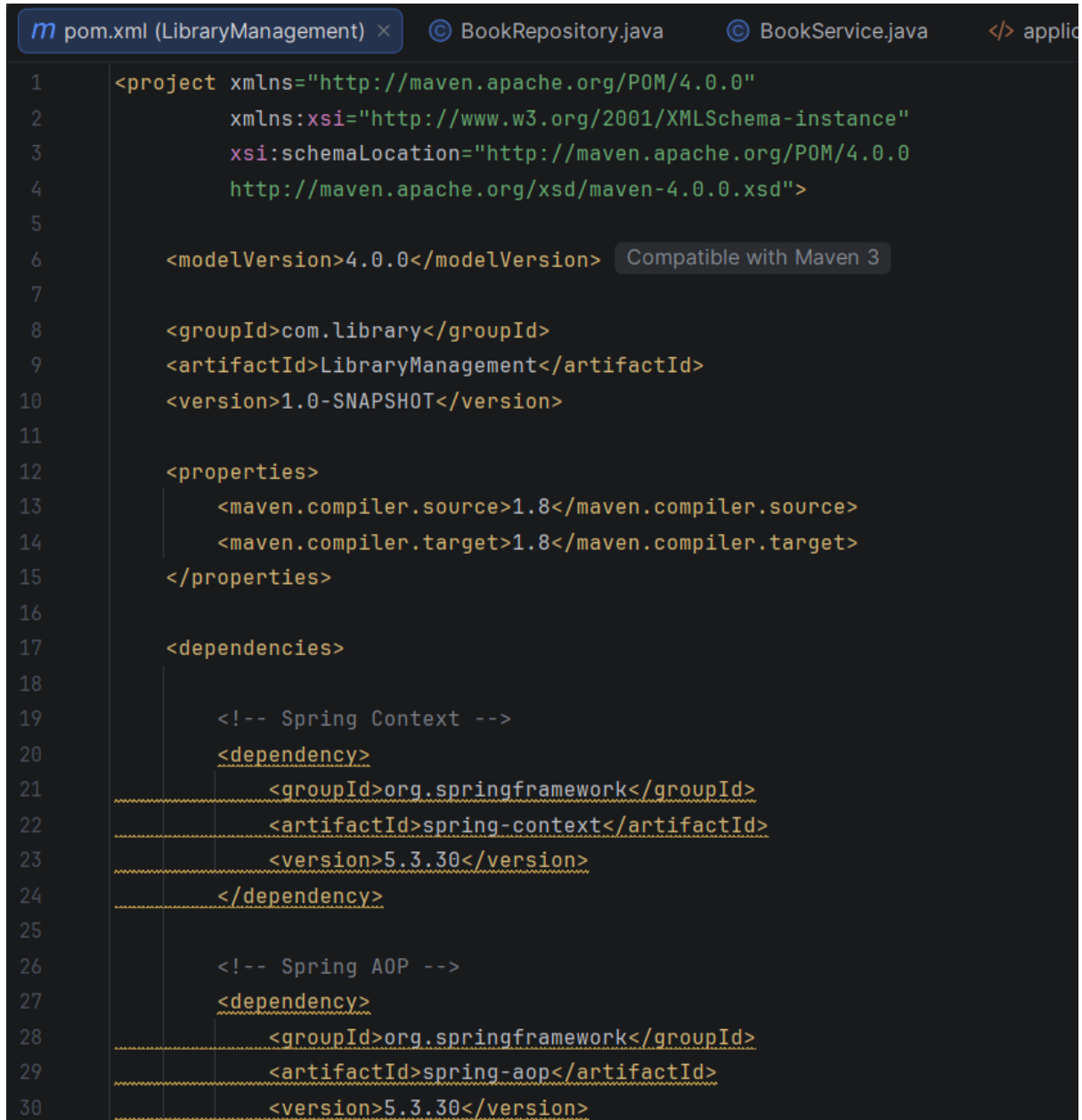


WEEK 3 Spring Core_Maven

Exercise 1: Configuring a Basic Spring Application



```
1 <project xmlns="http://maven.apache.org/POM/4.0.0"
2       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3       xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
4       http://maven.apache.org/xsd/maven-4.0.0.xsd">
5
6     <modelVersion>4.0.0</modelVersion> Compatible with Maven 3
7
8     <groupId>com.library</groupId>
9     <artifactId>LibraryManagement</artifactId>
10    <version>1.0-SNAPSHOT</version>
11
12    <properties>
13        <maven.compiler.source>1.8</maven.compiler.source>
14        <maven.compiler.target>1.8</maven.compiler.target>
15    </properties>
16
17    <dependencies>
18
19        <!-- Spring Context -->
20        <dependency>
21            <groupId>org.springframework</groupId>
22            <artifactId>spring-context</artifactId>
23            <version>5.3.30</version>
24        </dependency>
25
26        <!-- Spring AOP -->
27        <dependency>
28            <groupId>org.springframework</groupId>
29            <artifactId>spring-aop</artifactId>
30            <version>5.3.30</version>
```

Pom.xml

BookRepository.java

```
package com.library.repository;

public class BookRepository {

    public void display(){
        System.out.println("BookRepo is working");}

}
```

BookService.java

```
package com.library.service;

public class BookService {
    public void display(){
        System.out.println("BookService is Working");
    }
}
```

```
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <beans xmlns="http://www.springframework.org/schema/beans"
4       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5       xsi:schemaLocation="
6         http://www.springframework.org/schema/beans
7         https://www.springframework.org/schema/beans/spring-beans.xsd">
8
9     <bean id="bookService"
10         class="com.library.service.BookService"/>
11
12     <bean id="bookRepository"
13         class="com.library.repository.BookRepository"/>
14
15 </beans>
```

ApplicationContext.xml

```
package com.library;
import com.library.service.BookService;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class LibraryManagementApplication {

    public static void main(String[] args) {

        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");

        BookService service = (BookService) context.getBean("bookService");

        service.display();
    }
}
```

Main Function(LibraryManagementApplication)

Exercise 2: Implementing Dependency Injection

BookService.java

```
package com.library.service;
```

```
import com.library.repository.BookRepository;
```

```
public class BookService {
```

```
    private BookRepository bookRepository;
```

```
    // Setter method for Dependency Injection
```

```
    public void setBookRepository(BookRepository bookRepository) {
```

```
        this.bookRepository = bookRepository;
```

```
    }
```

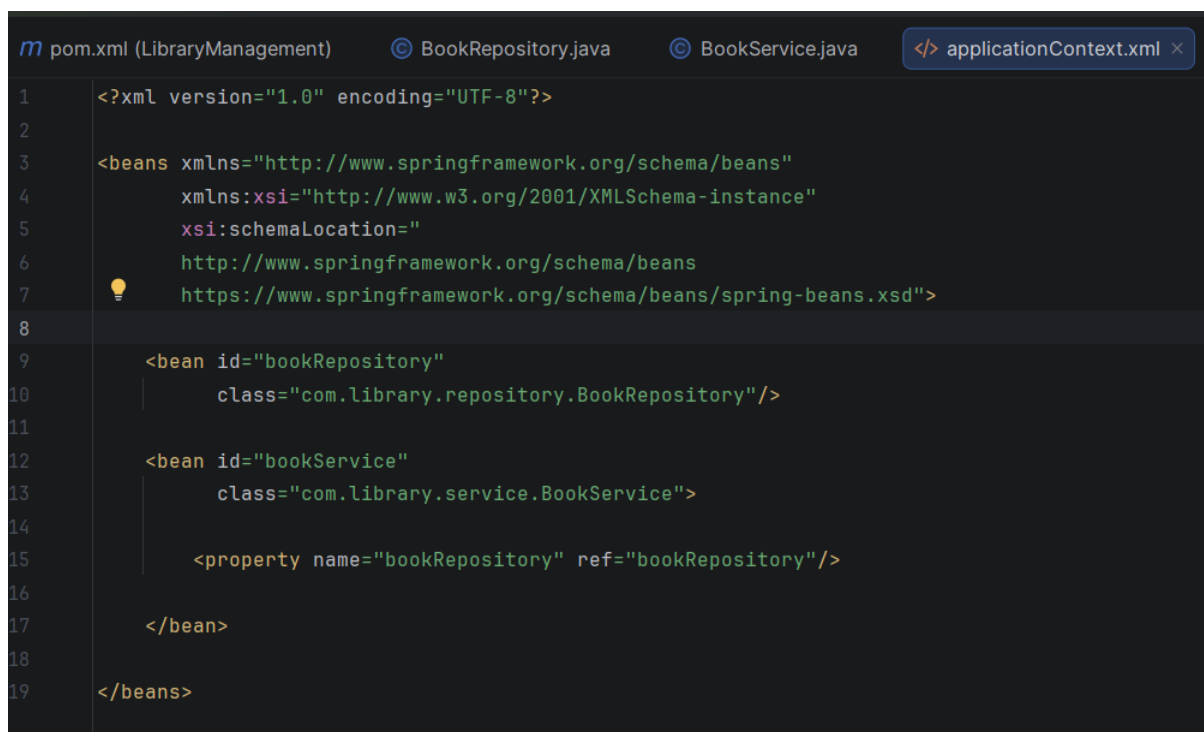
```
    public void display() {
```

```
        System.out.println("BookService is working...");
```

```
        bookRepository.display();
```

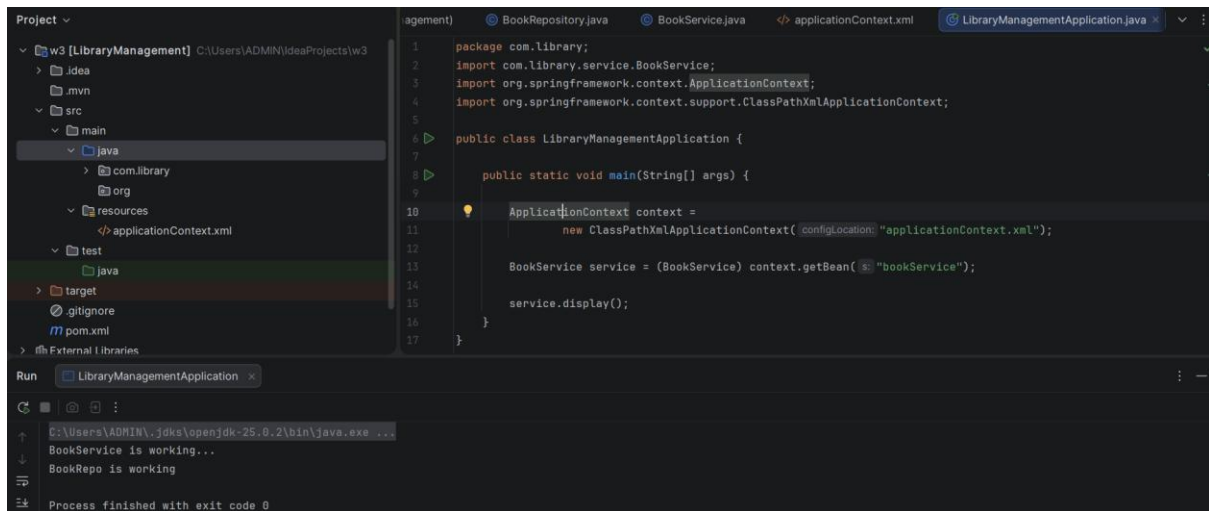
```
    }
```

```
}
```



```
m pom.xml (LibraryManagement)  BookRepository.java  BookService.java  </> applicationContext.xml x
1  <?xml version="1.0" encoding="UTF-8"?>
2
3  <beans xmlns="http://www.springframework.org/schema/beans"
4      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5      xsi:schemaLocation="
6          http://www.springframework.org/schema/beans
7          https://www.springframework.org/schema/beans/spring-beans.xsd">
8
9      <bean id="bookRepository"
10         class="com.library.repository.BookRepository"/>
11
12     <bean id="bookService"
13         class="com.library.service.BookService">
14
15         <property name="bookRepository" ref="bookRepository"/>
16
17     </bean>
18
19 </beans>
```

ApplicationContext.xml



LibraryManagemntApplication.java